

# **Optimization applied II: consumption-saving problem**

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# The simplest consumption-saving problem:

Let's assume that agents:

- Have a CRRA utility function over consumption:

$$u(c) = \begin{cases} \frac{c^{1-\sigma}-1}{1-\sigma} & \text{if } \sigma \neq 1 \\ \log(c) & \text{if } \sigma = 1 \end{cases}$$

- Live for 2 periods (young and old)
- Maximize the “lifetime” utility
- Receive income  $y$  in each period
- (Can) enter their life with some initial level of assets  $a_0 \neq 0$ .
- And cannot die with debt  $a_2 \geq 0$ !

## The simplest consumption-saving problem:

The Lifetime utility is:

$$U(c_1, c_2) = u(c_1) + \beta u(c_2) \quad (1)$$

Here  $\beta$  is the discount factor.

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Agent maximize equation (1) subject to constraints:

$$\max_{c_1, c_2, a_1, a_2} \underbrace{U(c_1, c_2)}_{\text{Lifetime utility function}}$$

$$c_1 + a_1 = a_0 R + y_1$$

$$c_2 + a_2 = a_1 R + y_2$$

Note:  $R = 1 + r$ . How to solve this problem?

## The simplest consumption-saving problem:

Let's look at the budget constraints. First note that  $a_2 = 0$  is optimal.

$$c_1 = a_0 R + y_1 - a_1$$

$$c_2 = a_1 R + y_2$$

Thus, the lifetime utility becomes a function of one variable!

$$\begin{aligned} U(c_1, c_2) &= u(c_1) + \beta u(c_2) \\ &= u(a_0 R + y_1 - a_1) + \beta u(a_1 R + y_2) \end{aligned}$$

# The simplest consumption-saving problem:

Note that we can define a function:

$$f(a_1) = u(a_0R + y_1 - a_1) + \beta u(a_1R + y_2)$$

where  $R, \beta, a_0, y_1, y_2$  are exogenous!

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For example assume:  $R = 1, \beta = 1, a_0 = 1, y = [3, 1]$ .

$$c_1 = a_0R + y_1 - a_1$$

$$c_2 = a_1R + y_2$$

Becomes:

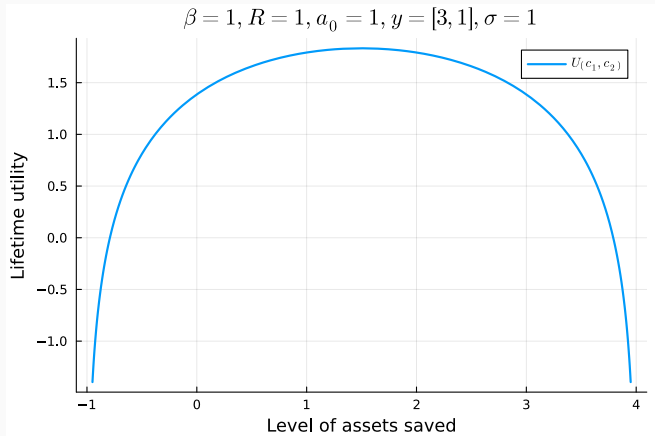
$$c_1 = 1 + 3 - a_1$$

$$c_2 = a_1 + 1$$

# The simplest consumption-saving problem:

$$f(a_1) = u(4 - a_1) + \beta u(a_1 + 1)$$

The lifetime utility assuming:  $R = 1, \beta = 1, a_0 = 1, y = [3, 1]$ .



How to find the maximum of  $f(a_1) = u(a_0R + y_1 - a_1) + \beta u(a_1R + y_2)$ ?

The first order condition is:

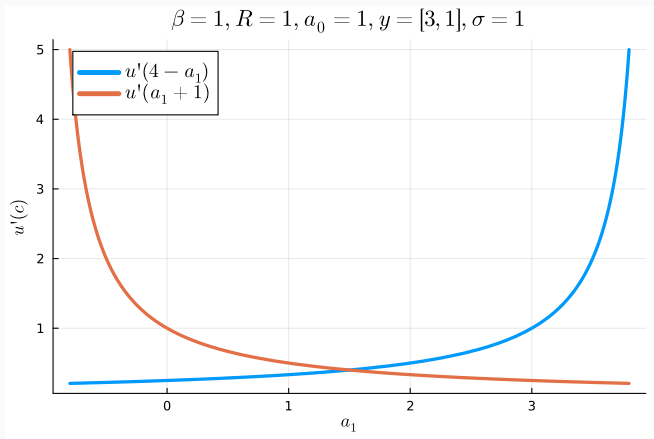
$$\frac{\partial f}{\partial a_1} = 0$$
$$u'(\underbrace{Ra_0 + y_1 - a_1}_{c_1}) = \beta Ru'(\underbrace{Ra_1 + y_2}_{c_2})$$

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## Let's add slightly more periods:

$$\max_{\{c_t, a_t\}_{t=1}^5} \underbrace{U(c_1, \dots, c_5)}_{\text{Lifetime utility}}$$

subject to:

$$c_1 + a_1 = a_0 R + y_1$$

$$c_2 + a_2 = a_1 R + y_2$$

$$c_3 + a_3 = a_2 R + y_3$$

$$c_4 + a_4 = a_3 R + y_4$$

$$c_5 + a_5 = a_4 R + y_5$$

Assumptions:

- Consumption cannot be negative (i.e.  $\forall_{t \in \{1, \dots, 5\}} : c_t \geq 0$ )
- The individual cannot die with debt (i.e.  $a_5 \geq 0$ )
- Borrowing and lending is possible in other periods (i.e.  $\forall_{t \in \{1, \dots, 4\}} : a_t \in \mathbb{R}$ )

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(3)

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If we had only two periods:

$$(c_1 - y_1) + \left( \frac{1}{R} \right) (c_2 + a_2 - y_2) = Ra_0.$$

Let's focus on the budget constraint:

$$\sum_{t=0}^4 \left(\frac{1}{R}\right)^t c_{t+1} - \sum_{t=0}^4 \left(\frac{1}{R}\right)^t y_{t+1} + \left(\frac{1}{R}\right)^4 a_5 = Ra_0$$

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**We get intertemporal budget constraint**

$$\sum_{t=0}^4 \left(\frac{1}{R}\right)^t c_{t+1} + \cancel{\left(\frac{1}{R}\right)^4 a_5} = \underbrace{\sum_{t=0}^4 \left(\frac{1}{R}\right)^t y_{t+1}}_{\text{Human wealth}} + \underbrace{Ra_0}_{\text{Initial financial wealth}}$$

## Intertemporal budget constraint

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- Instead of having many budget constraints we have one
- This is similar to a constraint you are all familiar with:

$$\sum_{t=0}^T p_t c_t \leq w$$

- **Observation:** interest rate  $r$  as an intertemporal price  $p_t$ :

$$p_t := \left(\frac{1}{1+r}\right)^t$$

- If individual has a **perfect foresight** and a **final planning horizon** distributing resources across periods is similar to distributing resources across various goods!

# Observation

- Timing of  $\{y_t\}$  does not matter for the optimal  $\{c_t\}$
- Two different paths of  $\{y_t\}$  resulting in the same  $H := \sum_{t=0}^T \left(\frac{1}{1+r}\right)^t y_t$  will:
  - lead to the same optimal consumption path,  $\{c_t\}$
  - lead to two different paths of  $\{a_t\}$
- This is because we were able to write a single intertemporal budget constraint
- No constraints on borrowing/lending allowed us to do so

## Optimization problem with intertemporal budget constraint:

$$\max_{\{c_t, a_t\}_{t=1}^5} \underbrace{U(c_1, \dots, c_5)}_{\text{Lifetime utility}}$$

subject to

$$\sum_{t=0}^4 \left(\frac{1}{R}\right)^t c_{t+1} + \left(\frac{1}{R}\right)^4 a_5 = \sum_{t=0}^4 \left(\frac{1}{R}\right)^t y_{t+1} + R a_0$$

$$a_5 \geq 0$$

$a_0$  given

$$\forall_{t \in \{1, \dots, 5\}} : c_t \geq 0$$

## Note on the last period savings

- Optimal  $a_5$  is zero
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- Things change with e.g. bequest motive

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Let's solve the model!

$$L = \sum_{t=0}^4 \beta^t u(c_{t+1})$$

$$- \lambda \left[ \sum_{t=0}^4 \left( \frac{1}{R} \right)^t c_{t+1} - \underbrace{\sum_{t=0}^4 \left( \frac{1}{R} \right)^t y_{t+1}}_{\text{Human wealth}} - \underbrace{Ra_0}_{\text{Initial financial wealth}} \right]$$

$$\frac{\partial L}{\partial c_{t+1}} : \beta^t u'(c_{t+1}) = \lambda \left( \frac{1}{R} \right)^t$$

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For  $t \in \{0, 1, 2\}$ :

$$u'(c_1) = \lambda$$

$$\beta u'(c_2) = \lambda \left( \frac{1}{R} \right)$$

$$\beta^2 u'(c_3) = \lambda \left( \frac{1}{R} \right)^2$$

From the above we get the Euler equation!

$$u'(c_1) = \beta R u'(c_2)$$

$$u'(c_2) = \beta R u'(c_3)$$

## The Euler equation:

$$u'(c_t) = R\beta u'(c_{t+1})$$

This can be also written as:

$$u'(Ra_{t-1} + y_t - a_t) = R\beta u'(Ra_t + y_{t+1} - a_{t+1})$$

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At the optimum agent is indifferent between:

- consuming a unit of resources today (and getting  $u'(c_t)$ )
- saving and consuming tomorrow (and getting  $(1 + r)\beta u'(c_{t+1})$ ).

Recall the CRRA utility function:

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}$$
$$u'(c) = c^{-\sigma}$$

Thus from:

$$u'(c_1) = R\beta u'(c_2)$$
$$u'(c_2) = R\beta u'(c_3)$$

We can get:

$$c_1^{-\sigma} = R\beta c_2^{-\sigma}$$
$$c_2^{-\sigma} = R\beta c_3^{-\sigma}$$

And finally:

$$c_2 = (R\beta)^{\frac{1}{\sigma}} c_1$$
$$c_3 = (R\beta)^{\frac{2}{\sigma}} c_1$$

## The Euler equation:

$$c_{t+1} = (R\beta)^{\frac{1}{\sigma}} c_t$$
$$\frac{c_{t+1}}{c_t} = (R\beta)^{\frac{1}{\sigma}}$$

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- The slope of the consumption  $\{c_t\}$  depends only on  $\beta$  vs.  $1 + r$

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- If  $\beta(1 + r) < 1$ , we have  $c_{t+1} < c_t$
- Forces: Impatience vs. returns on savings

## Two motives for savings

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- When  $\beta(1 + r) \neq 1$

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- In place even if  $\beta(1 + r) = 1$

2. **Intertemporal** motive

- When  $\beta(1 + r) \neq 1$
- Present even when  $y_t$  is constant over the lifecycle

# Solving the problem

We again can define a lifetime utility function:

$$\begin{aligned} f(a_1, \dots, a_4) = & u(\underbrace{a_0 R + y_1 - a_1}_{c_1}) + \beta u(\underbrace{a_1 R + y_2 - a_2}_{c_2}) + \beta^2 u(\underbrace{a_2 R + y_3 - a_3}_{c_3}) \\ & + \beta^3 u(\underbrace{a_3 R + y_4 - a_4}_{c_4}) + \beta^4 u(\underbrace{a_4 R + y_5}_{c_5}) \end{aligned}$$

Your task is to write this objective function into Julia!